



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. ECE B

Course Code & Title: EC8651 Transmission Lines and RF Systems

Name of the Faculty member: Mr.D.Gopinath

Innovative Practice Description

- **Unit / Topic:** Unit III / Impedance Matching methods
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Team Assisted Individualization (TAI)
- **Justification:**

Team Assisted Individualization (TAI) is used to improve the academic behaviour of the students, to increase their Problematic knowledge more than traditional methods. There are three major impedance matching techniques are available. Impedance matching networks can be designed by analytical method and graphical method (Smith Chart). In analytical method, conventional design equations are used to design matching networks. So, TAI method can be used to enrich the analytic knowledge of the students.

- **Time Allotted for the Activity:** One Period (50 Minutes)

- **Details of the Implementation:**

At first the students are divided into groups of four or 5 members (10 Groups). Discussed the Team Assisted Individualization (TAI) activity to the students. TAI activity objectives, outcomes, rubrics are explained and provided problems to be solved.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PSO3
CO3	3	2	2	2	2	2	3	3

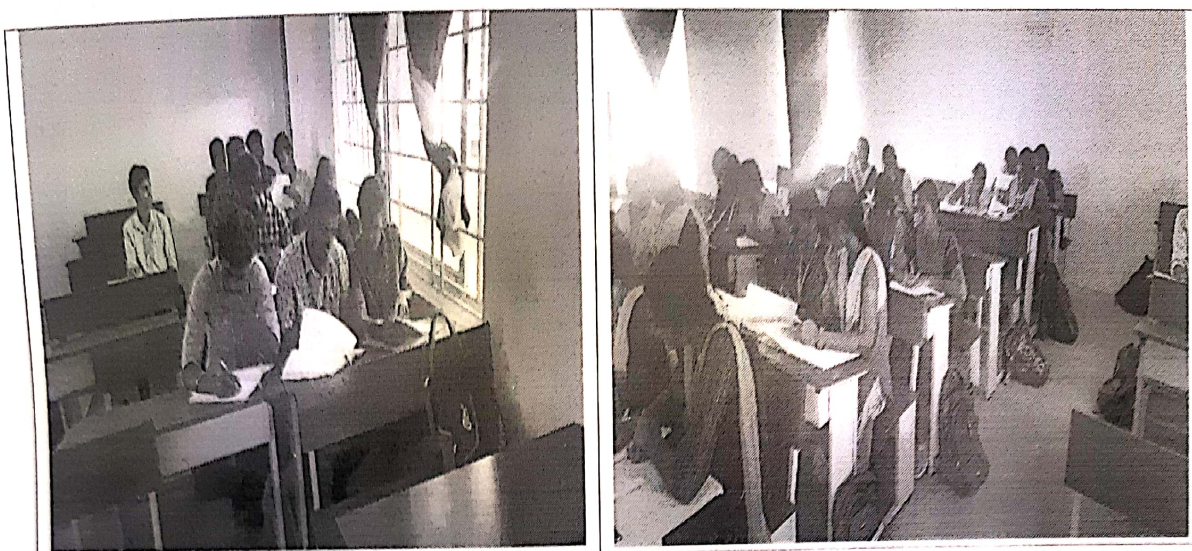
- **PO / PSO mapped:**

Innovative practice	PO4	PO8	PO9	PO10	PSO3
	2	2	2	3	3
Justification for correlation	The course outcome is	Identify tenets of the	A collaborative	The topics deals design	The concept of impedance

	moderately correlated since it involves the analysis of different impedance matching networks	professional code of ethics while doing activity-based learning. So, it is moderately correlated	activity TAI is planned which involves Implement the norms of practice (e.g. rules, roles, charters, agendas, etc.) of effective team work, to accomplish a goal. This course outcome is correlated moderately.	of impedance matching networks involves quarter wave transformer and single stub matching to complement writing and presentations. This task is accomplished through assignment therefore course outcome is correlated highly.	matching networks are required for design, analyze communication systems. So, the course outcome is highly correlated.
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• **Images / Screenshot of the practice:**





- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students feel that it was interesting. They learn how to communicate with team members and work together.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

Before taking this activity, students felt difficult to understand the problems of impedance matching networks. After completing this, they have rectified their doubts within that group. They have solved the impedance matching problems effectively in a group.

- ❖ **Challenges faced in implementation:**

Most of the student groups did the problems effectively. However, some of the student groups had different thoughts while solving the problems and also they have not communicated effectively with in that group. Due to that some minor errors are happened at the time of problem solving.

References:

- ❖ Slavin, R. E., Madden, N. A., & Leavey, M. (1984). Effects of team assisted individualization on the mathematics achievement of academically handicapped and nonhandicapped students. *Journal of Educational Psychology*, 76(5), 813–819. <https://doi.org/10.1037/0022-0663.76.5.813>
- ❖ John D Ryder, “Networks, lines and fields”, 2nd Edition, Prentice Hall India, 2015.
- ❖ <https://eric.ed.gov/?id=E10232852>
- ❖ <https://psycnet.apa.org/record/1985-10827-001>

Signature of Faculty Member

HOD



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Active Learning Activity- Unit-I- Transmission Line Theory

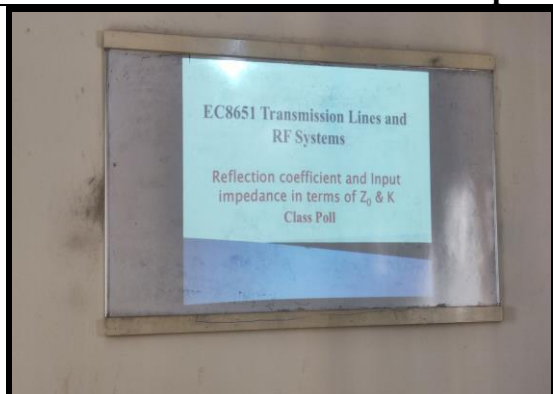
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Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Reflection coefficient and Input impedance in terms of Z_0 & K	Class Poll	21.03.2022 & 5 Minutes
Input and transfer impedance, Open and short circuited lines	Mind map	25.03.2022 & 5 Minutes

Reflection coefficient and Input impedance in terms of Z_0 & K - Class Poll

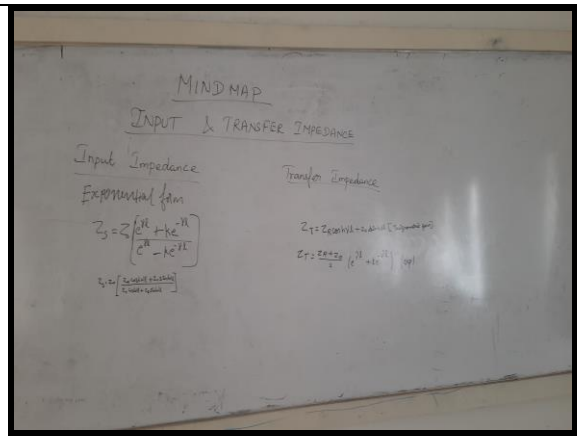
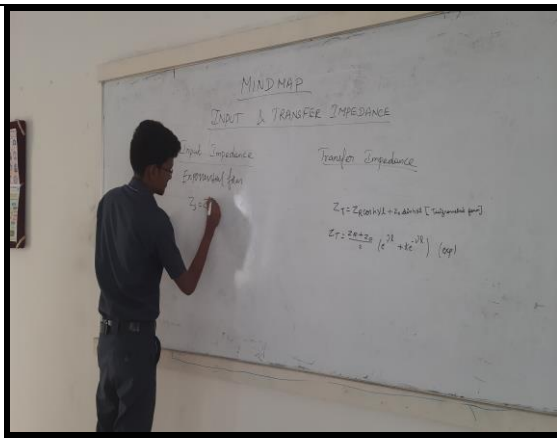
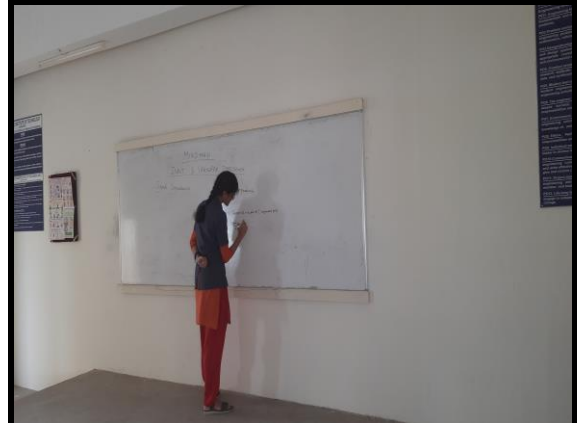
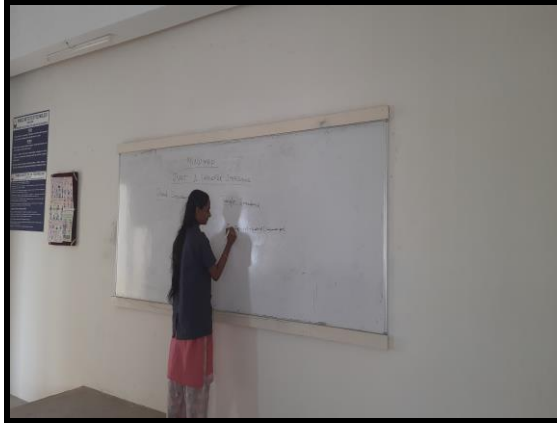




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Input and transfer impedance, Open and short circuited lines - Mind map





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Active Learning Activity- Unit-II- High Frequency Transmission Lines

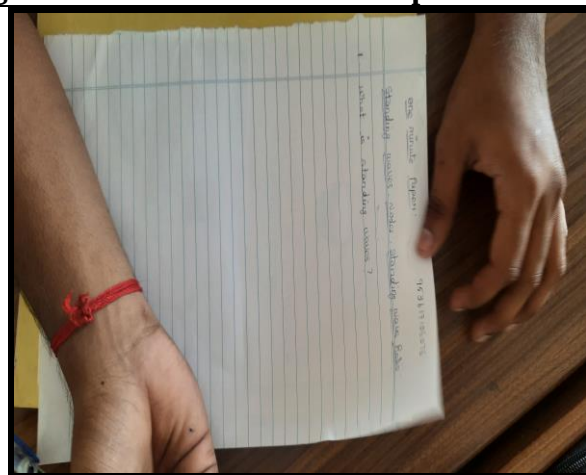
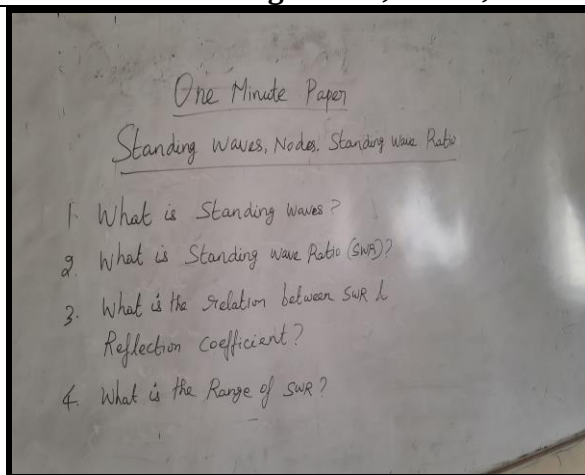
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Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Standing Waves, Nodes, Standing Wave Ratio	One Minute Paper	01.04.2022 & 5 Minutes
Measurement of VSWR and wavelength, Revision	Exit Slip	18.04.2022 & 5 Minutes

Standing Waves, Nodes, Standing Wave Ratio - One Minute Paper



Measurement of VSWR and wavelength- Exit Slip





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**Active Learning Activity- Unit-III- UNIT III IMPEDANCE MATCHING IN HIGH
FREQUENCY LINES**

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Name of the Topic	Name of the Activity	Date & Duration
Impedance matching: Quarter wave transformer and its applications	Brain Storming	19.4.2022 & 5 Minutes
Single stub matching using Smith chart	Team Assisted Individualization	13.5.2022 & 50 minutes

Impedance matching: Quarter wave transformer and its applications - Brain Storming



Signature of Faculty Member

4/7/22
HOD